



SARDAR PATEL UNIVERSITY
BCA SEMESTER-IV Examination 2025 (NC)
US04CBCA54 : Operating Systems

Date : 04/04/2025 (Friday)

TIME: 02:00 PM to 05:00 PM

TOTAL MARKS: 70

Notes

1. Figures to the right indicate full marks
2. All questions are compulsory

Q.1 MCQ**10**

- 1 A Program in execution is known as _____.
A. Process B. Files
C. Commands D. None
- 2 In monolithic structure, the OS divided into number of _____.
A. Program Control Block B. Process Control Block
C. Process Communication Block D. Program Communication Block
- 3 ____ instruction is use to convert user mode to kernel mode in monolithic structure.
A. TRAP B. PRAP
C. SCRAP D. GREP
- 4 ____ is not a valid Memory allocation algorithm.
A. Suit-fit B. Worse-fit
C. Next-fit D. First-fit
- 5 ____ is another name of enhanced second chance Page Replacement.
A. FIFO B. Optimal
C. LRU D. NRU
- 6 Swapping of processes requires _____.
A. Operating System B. Main Memory
C. Backing Store D. All of these
- 7 _____ is the solution of the bounded buffer.
A. Shared-Memory B. Data consistency
C. Virtual memory D. None
- 8 _____ buffer has fixed buffer size.
A. Bounded B. Unbounded
C. Shared D. None
- 9 _____ Option of ls command will sort output according to size of file.
A. -c B. -b
C. -S D. None
- 10 _____ command is use to merge multiple files.
A. merge B. paste
C. cp D. None

Q.2. Short Questions.(Attempt any ten)**20**

1. Define Operating system.
2. Differentiate between process and program.
3. List all layers of Layered structure of OS.

4. Explain External Fragmentation in short.
5. What is Compaction? For what purpose it will use?
6. Explain Next-fit memory allocation techniques.
7. Calculate Page faults using FIFO algorithm for following reference string: (Number of Frames = 3) Reference string = 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1
ANSWER: Page fault =15(FIFO)
8. When Race conditions arise?
9. Discuss advantages of Interview.
10. Explain Data Flow Diagram and list its types.
11. Explain who command in brief.
12. Explain cut command in short.

Q.3

- A. List out various types of OS. Explain Real Time Operating System. **05**
- B. Explain Round-Robin scheduling algorithm with example. **05**

OR

Q.3

- A. Explain Virtual machine in brief. **05**
- B. Explain SJF scheduling algorithm in brief. **05**

Q.4.

- A. Explain Memory allocation techniques in detail. **05**
- B. Explain FIFO page replacement algorithm with advantages and disadvantages. **05**

OR

- Q.4.** Define Page Fault. Explain page fault handling technique. **10**

Q.5

- A. What is LINUX? Explain basic features of LINUX Operating System. **05**
- B. Explain algorithm 3 for two-process solution. **05**

OR

- Q.5.** What do you mean by Deadlock? Explain all necessary conditions for occurrence of deadlock. **10**

Q.6.

- A. Explain following commands: a) date b) ls c) rmdir **05**
- B. Explain for loop in Linux. **05**

OR

- Q.6.** What is Shell script? Explain uses of Shell script and execution of shell script. **10**

_____ x _____